

The deeper you go into computer programming, the more likely it is that you will find copious amounts of C. Nearly every electronic device you touch, that has some kind of embedded microcontroller or microprocessor in all likelihood runs at least some code that was once stored in a .c file.

Hello World

```
• #include <stdio.h>
•
• int main(void) {
•     printf("Hello, world!\n");
•     return 0;
• }
```

Hello World is possibly one of the simplest programs you can write in any language, and C is possibly one of the more verbose ones. If you just want something that compiles – with warnings – and throws “Hello, world!” to the screen though, you can even get away with just the following (with some compilers):

```
• main() {
•     printf("Hello, world!\n");
• }
```

Sort

```
• void insertion_sort(int *arr, const size_t arr_len)
{
•     for (size_t i = 1; i < arr_len; i++) {
•         int elem = arr[i];
•         size_t j = i;
•         for (; j > 0 && elem < arr[j - 1]; j--) {
•             arr[j] = arr[j - 1];
•         }
•         arr[j] = elem;
•     }
• }
```

Insertion sort in C isn’t exactly complex. Here in the above example we are the newer syntax which allows us to define variable as they are used rather than in the beginning of the function. This has been supported since C99.